Matteo Romilio Pasqual

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"In the middle of difficulty lies opportunity." – Albert Einstein

Education

Politecnico di Milano M.Sc. in Computer Science and Engineering - Artificial intelligence *Sep 2024 – Today*

Politecnico di Milano *Sep 2021 – Jul 2024*
B.Sc. in Computer Science and Engineering (polimi.it) Grade: 109/110

Liceo Scientifico "Leonardo da Vinci" *Sep 2016 – Jul 2021*
Diploma di Liceo Scientifico - Scienze Applicate (liceogallarate.edu.it) Grade: 96/100

Projects and Research

SHRED-Based Sensor Reconstruction in High-Dimensional Dynamical Systems with Missing and Irregular Measurements *Ongoing Thesis - 2026*

- Investigating SHRED-based architectures for state reconstruction in high-dimensional dynamical systems under missing, asynchronous, and unreliable sensor measurements
- Studying the impact of incomplete temporal coverage on recurrent and attention-based encoders within the SHRED framework developing and benchmarking strategies for handling missing data, including imputation methods and masking mechanisms for sequential models

Multi-agent System Leonardo Drone Contest 2025 *Project - 2025*

- Participated in a national robotics competition (Leonardo Drone Contest)
- Designed a multi-agent system for collaborative navigation and dynamic task allocation in unknown indoor environments, developing decentralized decision-making strategies, inter-agent communication protocols, and computer vision modules for object detection and pose estimation
- Strengthened skills in teamwork, robustness engineering, and problem-solving under uncertainty

Course Projects *Project - 2025*

- ANNDL course project: multivariate time-series classification using neural networks (GitHub)
- ANNDL course project: biomedical image classification using deep learning methods (GitHub)
- MMMIP course project: image processing for denoising, inpainting, and anomaly detection (GitHub)
- OLA course project: pricing under budget constraints using online learning methods (GitHub)

Kolmogorov-Arnold Networks (KAN) lecture *Research - 2024*

- Delivered a lecture on KANs, a neural architecture inspired by the Kolmogorov–Arnold representation theorem, as part of the Numerical Analysis for Machine Learning course
- Presented the theoretical foundations of KANs and their differences from MLP, focusing on spline-based learnable activation functions, training methodology, sparsification, pruning, and symbolic regression techniques, highlighting advantages in interpretability and function approximation
- Achieved top evaluation (30/30) and selected for inclusion in future NAML lectures at PoliMi
- Developed supporting materials and implementation examples (GitHub)

Non-intrusive Monitoring of Daily Habits in Older Adults: A Chaos Game Representation-Based Approach *Research - 2024*

- Co-authored a paper accepted at UCAmI 2025 – 17th International Conference on Ubiquitous Computing and Ambient Intelligence (Paper)
- Developed a novel framework using Chaos Game Representation (CGR) to transform time-series sensor data into fractal-like images for behavioural analysis, enabling classification of daily routines.
- Demonstrated effective anomaly detection and clustering of behavioural patterns on both synthetic and real-world multi-resident datasets

Experience

University Multichanche Tutor *Milano, IT*
Politecnico di Milano *Sep 2025 – Jul 2026*

- Tutored university students with the sponsorship of the Multichance Team
- Provided personalized academic support and study assistance

University Peer-to-peer Tutor *Milano, IT*
Politecnico di Milano *Sep 2022 – Jul 2024*

- Tutored university students with the sponsorship of Univeristy
- Provided personalized academic support and study assistance

Languages

- **Italian**: native
- **English**: C1

Skills

- **Programming & Software Dev.**: Python, C/C++, Java
- **Databases Technologies**: SQL, MongoDB, Spark, Neo4j, Zilliz-Milvus
- **OS & Middelware**: ROS, ROS2, Linux, Docker and VMs
- **Other Technologies**: Matlab, VHDL/Vivado

Awards

Best Freshmen Award: top 5% freshmen of PoliMi 2023

Volounteering

Volounteer at Oratorio di Albizzate Set 2016 - Aug 2021
Albizzate